2. Random Walk on Networks

April 15, 2026

1 (a) Graph Construction

We generate an Erdős–R'enyi (ER) graph with $n = 900$ and edge probability $p = 0.015$. The resulting graph is connected, with 900 nodes and 6121 edges. The mean degree is 13.500.

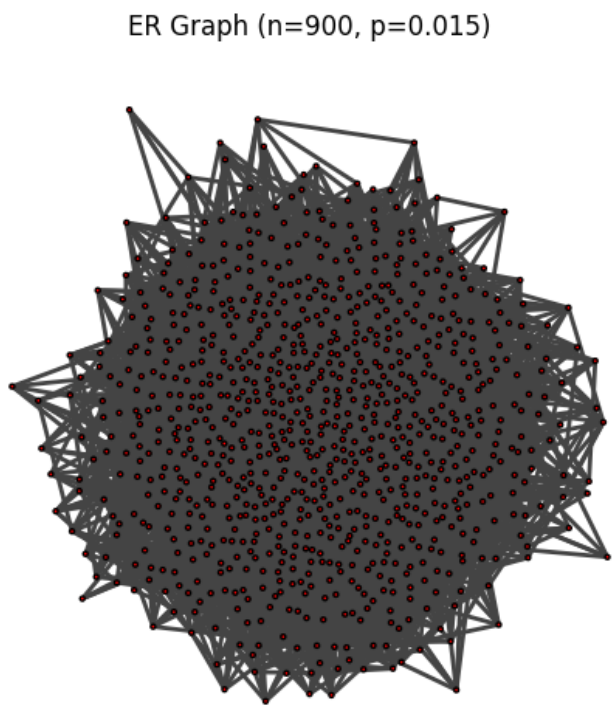

Figure 1: Visualization of the ER graph with $n = 900$ and $p = 0.015$.

1 (b) Random Walk Behavior

We let a random walker start from a randomly chosen node and measure the average distance $\langle s(t) \rangle$ from the starting node together with its variance as a function of time t.

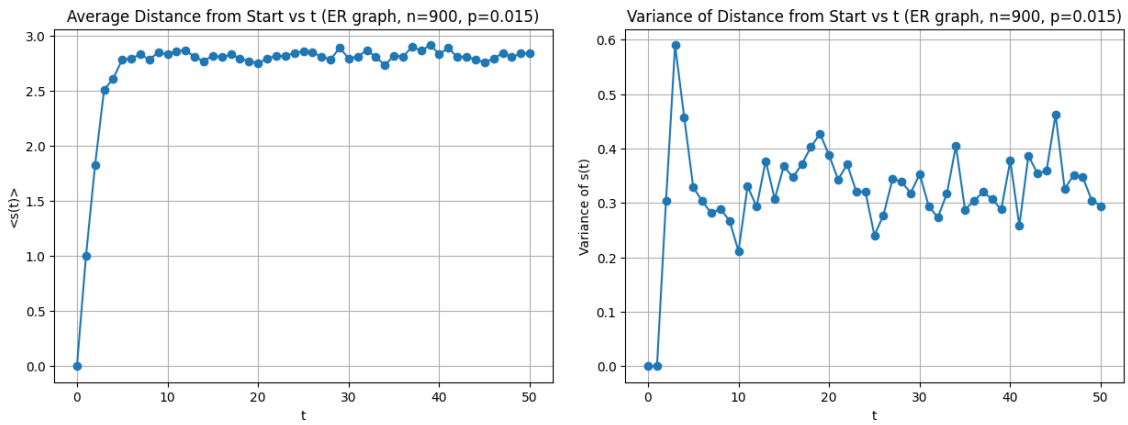

Figure 2: Average distance (left) and variance (right) of the distance from the starting node as a function of t for the ER graph.

The average distance increases rapidly at small t and then saturates around 2.8–2.9. The variance also rises quickly and then fluctuates around a roughly constant level. This indicates that the random walk reaches a stationary regime after a small number of steps, which is consistent with the short path lengths in the ER graph.

1 (c) Endpoint Degree Distribution

We compare the degree distribution of all nodes in the ER graph with the degree distribution of the nodes reached at the end of the random walk.

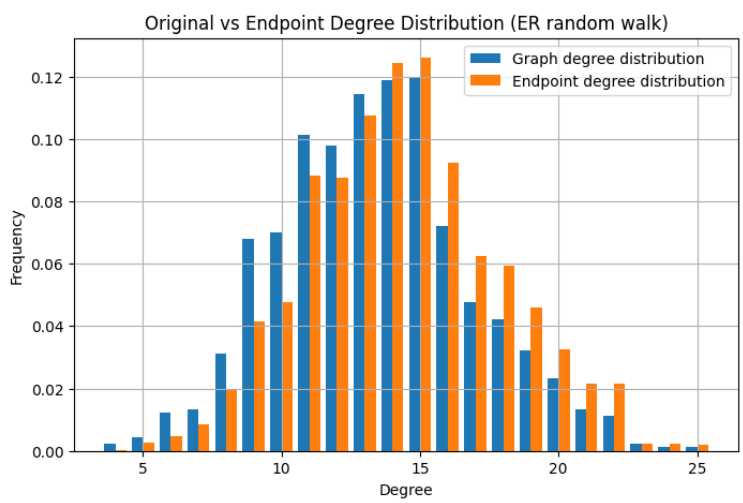

Figure 3: Original degree distribution and endpoint degree distribution for the ER graph.

The endpoint distribution is slightly shifted toward larger degrees. In particular, the mean degree of all nodes is 13.431, while the mean degree of endpoint nodes is 14.365.

This shows that higher-degree nodes are more likely to be visited by the random walk, although the effect is relatively mild because the ER graph has a narrow degree distribution.

1 (d) Effect of Graph Size

We repeat the random walk experiment on a larger ER graph with $n = 9000$ and $p = 0.015$.

The graph is connected, with mean degree 135.184 and diameter 3.

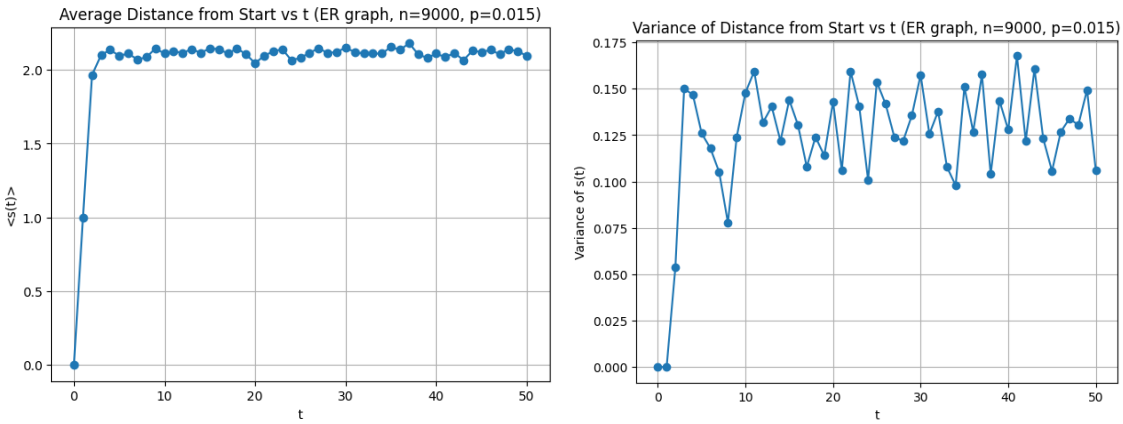

Figure 4: Average distance (left) and variance (right) for the ER graph with $n = 9000$.

Compared to the smaller graph ($n = 900$), the average distance saturates faster and at a lower value (around 2.1 vs 2.8). The variance is also smaller.

This is because the larger graph has a much higher average degree, making it more densely connected and reducing typical path lengths. As a result, the random walk mixes faster and reaches the stationary regime in fewer steps.

2 (a) Graph Construction

We generate a preferential attachment graph with $n = 900$ and $m = 1$.

The graph is connected, with 900 nodes and 899 edges, and mean degree 1.998.

Figure 5: Visualization of the preferential attachment graph with $n = 900$ and $m = 1$.

2 (b) Random Walk Behavior

We let a random walker start from a randomly chosen node and measure the average distance $\langle s(t) \rangle$ from the starting node together with its variance as a function of time t.

The graph is connected, with mean degree 1.998 and diameter 20.

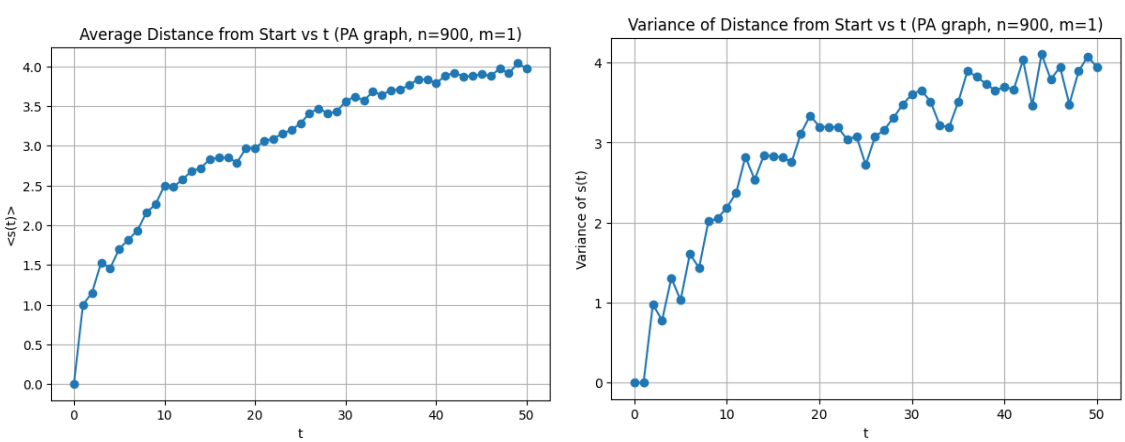

Figure 6: Average distance (left) and variance (right) of the distance from the starting node as a function of t for the preferential attachment graph.

Unlike the ER graph, the average distance continues to increase over time instead of quickly saturating, and the variance also keeps growing. This indicates slower mixing on the PA graph,

which is consistent with its tree-like structure and much larger diameter.

2 (c) Endpoint Degree Distribution

We measure the degree distribution of nodes reached at the end of the random walk and compare it with the degree distribution of the graph.

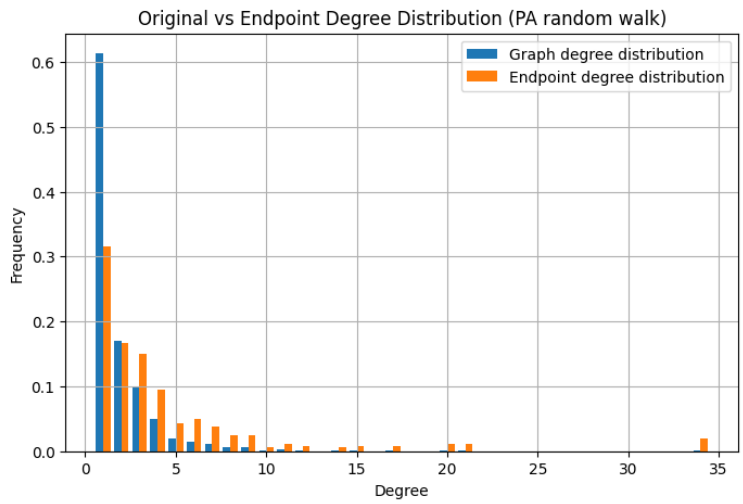

Figure 7: Original degree distribution and endpoint degree distribution for the preferential attachment graph.

The endpoint distribution is strongly shifted toward higher degrees. The mean degree increases from 1.998 (all nodes) to 4.456 (visited nodes).

This bias is much stronger than in the ER graph because the PA graph has a heavy-tailed degree distribution. High-degree hubs attract the random walk more frequently, making them significantly more likely to be visited.

(2d) Effect of Graph Size

We repeat the random walk experiment on preferential attachment graphs with $n = 90$, 900, and 9000.

The corresponding mean degrees are 1.978, 1.998, and 2.000, while the diameters are 9, 22, and 29, respectively.

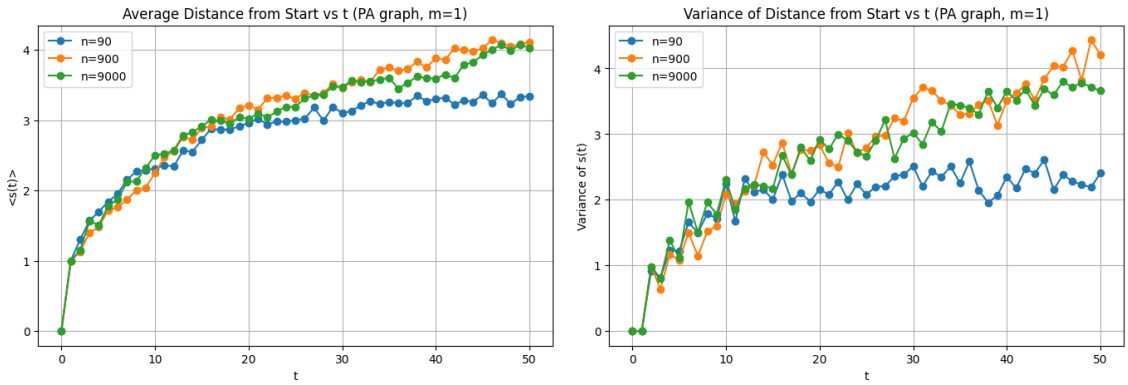

Figure 8: Average distance (left) and variance (right) for preferential attachment graphs of different sizes.

As the graph size increases, both the average distance and the variance tend to become larger. The $n = 90$ graph reaches a lower distance range, while the $n = 900$ and $n = 9000$ graphs continue growing to larger values. A similar trend is observed for the variance, where larger graphs show broader fluctuations in distance.

This behavior is consistent with the increase in graph diameter from 9 to 22 to 29, while the mean degree stays nearly constant at about 2. Since larger PA graphs remain sparse but become more extended, the random walker can reach nodes at larger distances and the spread of distances also becomes wider.

3 (a) Visit Probability vs Degree

We analyze the relationship between node visit probability and node degree.

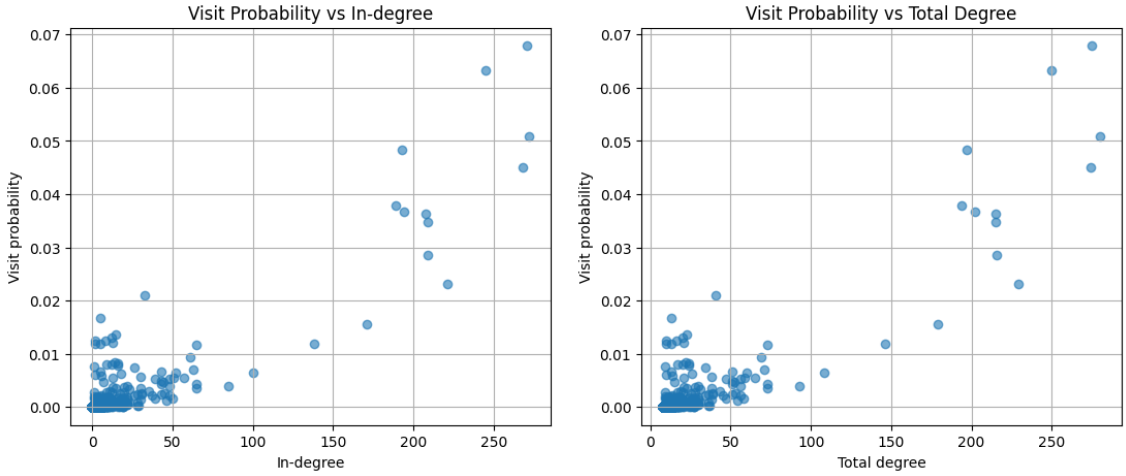

Figure 9: Visit probability as a function of in-degree (left) and total degree (right).

The results show a strong positive correlation between visit probability and node degree. Nodes with higher in-degree (and total degree) are significantly more likely to be visited during the random walk.

In particular, the most frequently visited nodes all have very large in-degrees (e.g., above 200),

while low-degree nodes have near-zero visit probability. This indicates that the random walk is heavily biased toward high-degree nodes.

This behavior is consistent with theory: in a random walk on a graph, the stationary distribution is proportional to node degree, i.e.,

$$\pi_i \propto k_i.$$

Therefore, high-degree nodes act as hubs that attract the random walker more frequently.

3 (b) Visit Probability with Teleportation

We introduce teleportation with parameter $\alpha = 0.2$ and analyze the relationship between visit probability and node degree.

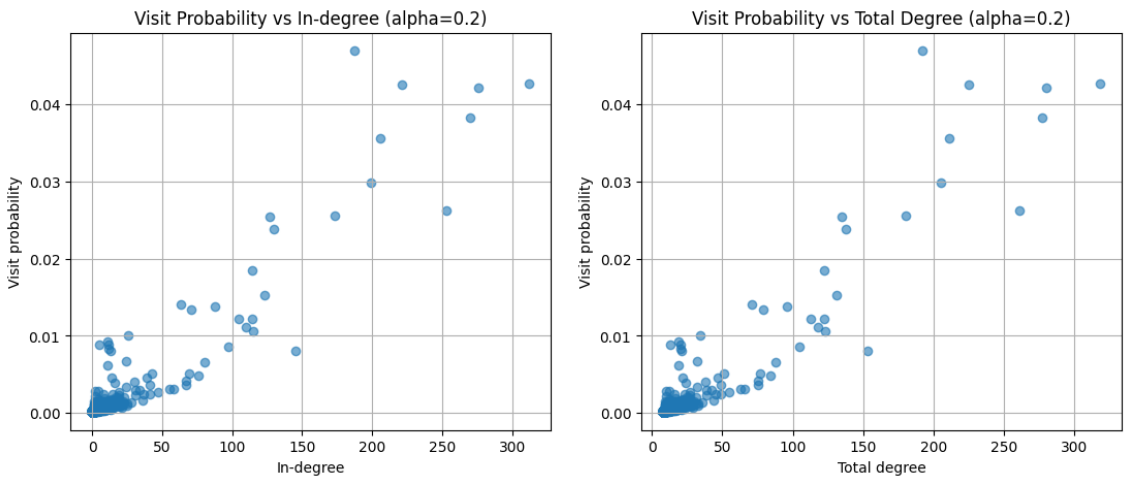

Figure 10: Visit probability as a function of in-degree (left) and total degree (right) with teleportation ($\alpha = 0.2$).

The visit probability still shows a positive correlation with node degree, but the dependence is weaker compared to the non-teleportation case. High-degree nodes remain more likely to be visited, but low-degree nodes now receive non-negligible probability due to random jumps.

Compared with part (3a), the distribution is less concentrated on hubs. Teleportation allows the random walker to explore the graph more uniformly, reducing the dominance of high-degree nodes.

This behavior is consistent with the PageRank model, where the stationary distribution is a mixture of degree-based random walk and uniform teleportation:

$$\pi = (1-\alpha)P^T\pi + \alpha\frac{1}{n}.$$

4 (a) Personalized Teleportation Based on PageRank

In this experiment, the teleportation probability is proportional to each node's PageRank, instead of being uniform as in the standard random walk. The teleportation parameter is set to $\alpha = 0.2$.

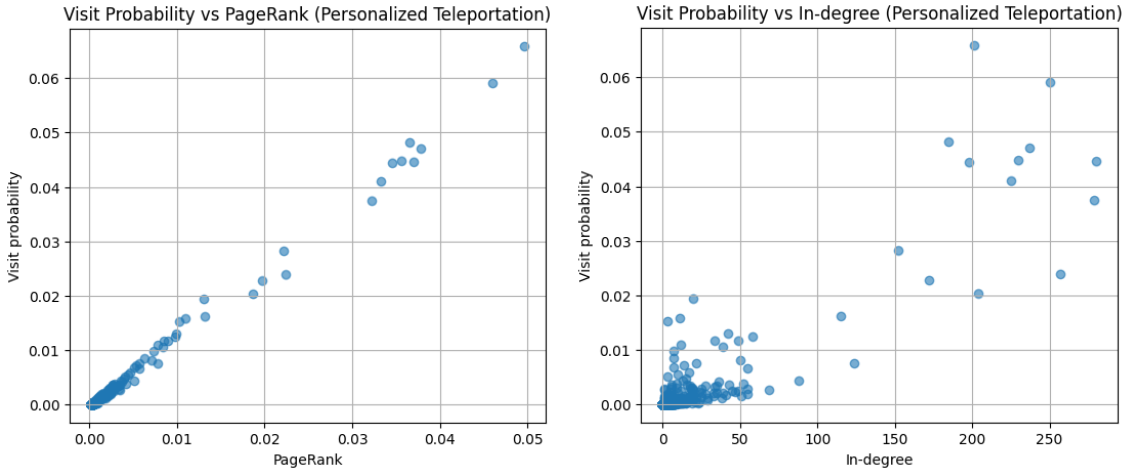

Figure 11: Visit probability vs PageRank (left) and in-degree (right) under personalized teleportation.

Compared with 3(a), the correlation between visit probability and in-degree becomes weaker, while the relationship with PageRank becomes significantly stronger. The left plot shows an almost linear trend between visit probability and PageRank, indicating that the random walk closely follows the personalized PageRank distribution.

This behavior occurs because teleportation directly biases the walk toward high-PageRank nodes. As a result, important nodes are reinforced both by the network structure and by the teleportation mechanism. In contrast, in 3(a), the visit probability is mainly determined by degree through local transitions.

Therefore, personalized teleportation shifts the importance measure from degree-based to PageRank-based, and produces a distribution that more accurately reflects global node importance.

4 (b) Teleportation to Two Median-PageRank Nodes

We select two nodes with median PageRank values and repeat the personalized random walk, but now teleportation lands only on these two nodes with probabilities 1/2 and 1/2.

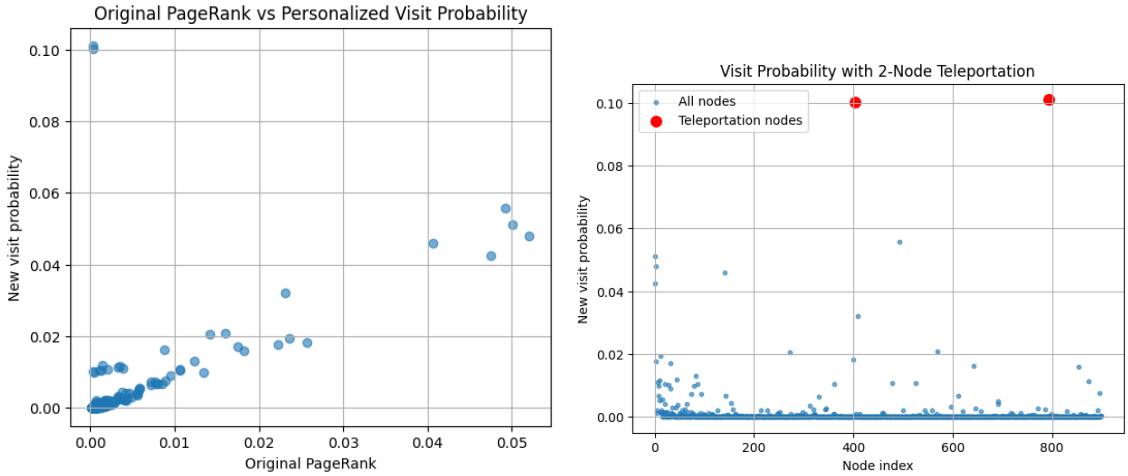

Figure 12: Original PageRank vs new visit probability (left), and visit probability under 2-node teleportation (right).

The two selected nodes are node 794 and node 403, each with original PageRank 0.000274. After restricting teleportation to these two nodes, their visit probabilities increase dramatically to 0.100959 and 0.100064, respectively.

Compared with part 4(a), the distribution is no longer aligned with the original PageRank. Instead, the walk becomes highly concentrated around the two teleportation nodes. This is clearly shown in the left plot, where two strong outliers appear: nodes with very small original PageRank now have the largest visit probabilities.

Therefore, restricting teleportation to only two nodes significantly changes the PageRank distribution. The ranking becomes highly localized, and the two selected nodes become dominant even though they were not important in the original PageRank. This shows that teleportation can strongly reshape node importance and override the original global ranking.

4 (c) Visiting Probability vs Personalized PageRank

We compare the visiting probability obtained from random walk with teleportation and the personalized PageRank distribution.

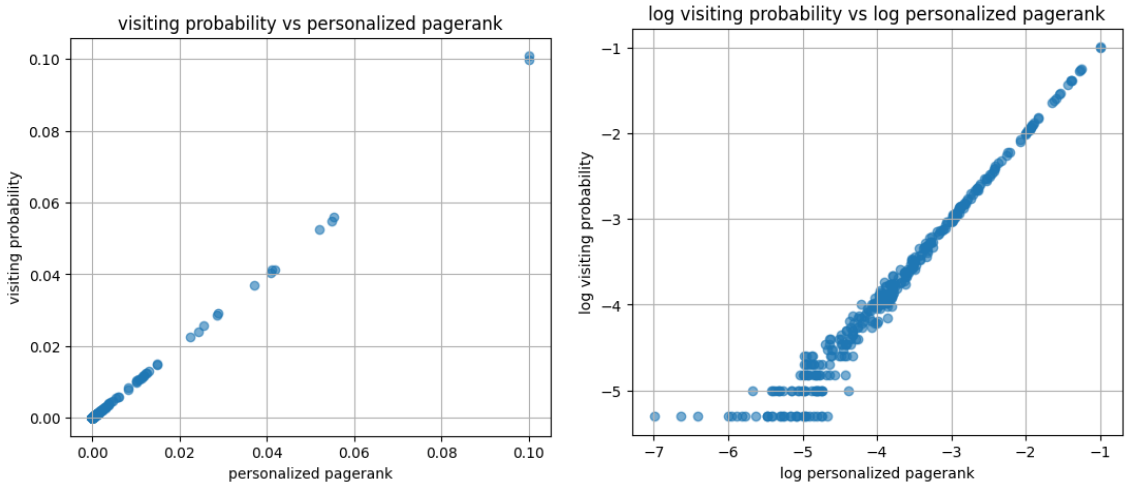

Figure 13: Visiting probability vs personalized PageRank (left), and their log-log relationship (right).

The results show an almost perfect linear relationship between visiting probability and personalized PageRank. The Pearson correlation coefficient is 1.0000, indicating that the two distributions are nearly identical.

In addition, the log-log plot also exhibits a clear linear trend, further confirming the proportional relationship between visiting probability and personalized PageRank.

This demonstrates that the random walk with teleportation converges exactly to the personalized PageRank distribution. Nodes with higher personalized PageRank are visited more frequently, and the visiting probability is effectively determined by the personalized PageRank values.

Therefore, personalized PageRank accurately characterizes the stationary distribution of the random walk under biased teleportation.